Unsupervised Multiview Graph Contrastive Feature Learning for Hyperspectral Image Classification

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***Abstract*— As a popular deep learning (DL) algorithm, graph neural network (GNN) has been widely used in hyperspectral image (HSI) classification. However, most of the GNN-based classification algorithms are concentrated in the field of semisupervision, which heavily relies on the quantity and quality of samples. To solve this problem, we propose an unsupervised multiview graph contrastive (UMGC) feature learning algorithm to explore the deep semantic features of HSIs without being constrained by samples. First, we construct multiview adjacency matrixes from spatial and spectral directions. Second, the adaptive data augmentation method is used to selectively enhance the topology and attribute structure of the graph. Thereafter, features are extracted by using a contrastive loss to maximize the similarity between the two views. Finally, we tested the model's performance based on multiple evaluation methods. Experimental results on three publicly available hyperspectral datasets show that the proposed UMGC can have better classification performance compared with other state-of-the-art unsupervised feature extraction (FE) methods.**

***Index Terms*— Contrastive learning (CL), graph convolutional network (GCN), hyperspectral image (HSI) classification, unsupervised feature learning.**

NOMENCLATURE

Notation	Definition
$\mathbf{X}$	HSI.
S, $\mathbf{F}$	Superpixel, superpixel features.
$\mathbf{A}^{\text{spa}}$, $\mathbf{A}^{\text{spe}}$	Spatial adjacency matrix, spectral adjacency matrix.
G^{spa}, G^{spe}	Spatial graph, spectral graph.
$\mathbf{M}$	Segment matrix.
$\mathbf{Q}$	Association matrix.
$\mathbf{W}$	Weight of node centrality.
$\mathbf{E}$	Edge_index of the adjacency matrix.
V	Node of graph.

Manuscript received 20 May 2024; revised 4 July 2024; accepted 16 July 2024. Date of publication 22 July 2024; date of current version 1 August 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62222116, Grant 62171417, and Grant 62071438. *(Corresponding author: Yanni Dong.)*

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Digital Object Identifier 10.1109/TGRS.2024.3431680

I. INTRODUCTION

WITH the prosperous development of spectral imaging technology and the sensor platform, hyperspectral images (HSIs) are playing an increasingly important role and have been widely developed in the fields of agricultural assessment, environmental monitoring, mineral identification, and land cover classification [1], [2], [3], [4]. These applications rely on pixel-level HSI classification technology. HSI can obtain hundreds of continuous spectral curves from the visible region (0.4–0.7 μm) to the short-wave infrared region (0.7–2.4 μm), providing detailed spectral information [5], [6]. The enormous abundance of data can also result in data redundancy problems. How to extract effective information from HSI is an essential research direction [7]. Therefore, feature extraction (FE), which can significantly reduce the dimensionality of HSI, is proposed as a crucial preprocessing step to improve the separability of various categories in classification tasks and enhance the performance of classification [8], [9].

In the early stages, numerous studies focused on reducing the dimensionality of HSI for FE, such as principal component analysis (PCA) [10], linear discriminant analysis (LDA) [11], nuclear fisher discriminant analysis (KFDA) [12], and sparse representation based on dictionary learning [13]. These studies also utilized classifiers, such as support vector machine (SVM) [14] and random forest (RF) [15], to classify low-dimensional features. However, the results obtained by these approaches are unsatisfactory because they only consider spectral features. The spatial features are not taken into account. In recent years, spatial features have been useful in improving the feature representation and classification accuracy of HSI [16], [17]. Benediktsson et al. [18] used extended morphological contours (EMPs) to incorporate spatial information through multiple morphological operations. Nevertheless, the above-mentioned traditional methods depend on experts' manual design features, and the shallow FE limits the accuracy of classification.

More recently, since deep learning (DL) has achieved remarkable revolutions in computer vision tasks [19], [20], [21], it has been widely applied in HSI classification because of its excellent ability to extract high-level semantic information [22], [23]. For example, Li et al. [24] used 3-D convolutional neural networks (CNNs) for accurate classification. This method does not rely on any preprocessing or postprocessing to effectively extract the deep spatial–spectral features. Guan and Lam [25] used a CNN-based contrastive

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learning (CL) framework to explore cross-domain information between the spatial and the spectral domains. Although the FE of HSI has been significantly improved by CNN [26], the CNN-based methods still exhibit some shortcomings. For example, convolution kernels have a fixed local receptive field, which means that they are unable to effectively capture long-range dependencies. In addition, deep CNNs require a substantial amount of training time due to their high computational complexity [27].

In recent years, graph convolutional networks (GCNs) have attracted increasing attention due to their ability to quickly perform convolution operations on the graph in non-Euclidean space and obtain long-range node information [28]. At present, two methods are primarily used to construct graphs from HSI data. The first method is used for graph-level classification [29], where the image is divided into multiple subgraphs using the pixels and its neighbor pixels, namely the patch, and each patch builds a subgraph. Subsequently, these subgraphs are divided into batches for training. Liu et al. [30] achieved excellent accuracy in graph-level classification by combining CNN and GCN through a supervised classification approach. The second method is used for node-level classification [29], where the entire image is divided into several superpixels using the simple linear iterative clustering (SLIC) [31], [32] approach. Numerous subsequent semisupervised algorithms inherited the idea [33], [34], [35]. Based on [34], our previous work proposed the fusion of weighted features of CNN and graph attention networks for HSI classification, which realized complementarity in classification details [36]. However, the aforementioned algorithms require a certain number and quantity of quality samples to optimize the parameters, which is expensive and time-consuming for HSI.

Additionally, most graph neural network (GNN) algorithms rely on the basic assumption that the original graph is authentic and can be treated as real information for model training. However, the construction of a graph in HSI is currently imperfect. In [33], [34], [35], and [36], the spatial information of first-order neighbor superpixels is relied on, when constructing adjacency matrix $\mathbf{A}$. The insufficient utilization of HSI spatial information restricts the extraction of local information. More crucially, existing methods fail to utilize the abundant spectral information in HSI when constructing adjacency matrices, neglecting the connectivity between spectrally similar superpixels. This may lead to the loss of long-range information in HSI. Consequently, the initial graphs constructed inevitably suffer from uncertainties, errors, and missing connections.

Furthermore, CL, which is one of the self-supervised representative learning methods, is widely used for unsupervised FE [25], [37], [38]. Graph augmentation is an important aspect of CL. Existing graph data augmentation methods mainly fall into four categories: feature masking [39], node dropping [40], edge perturbation [41], and subgraph sampling [42]. Most graph algorithms only obtain two views by randomly augmenting the input graph [39], [43], [44]. However, random augmentation will lose critical connections and features in the graph, which may force the model to learn unimportant nodes and connections [45]. Zhu et al. [45] proposed adaptive augmentation in three directions: degree, eigenvector, and pagerank. However, regarding node attribute-level augmentation, it employs a relatively simplistic approach by randomly adding salt-and-pepper noise to node features using a Bernoulli distribution [46], which fails to fully achieve the goal of adaptive augmentation. Consequently, we aspire to design a graph-structured adaptive data augmentation module (ADAM), aiming to further enhance the FE capabilities of CL models.

In order to address the above problems, we propose an unsupervised multiview graph contrastive (UMGC) feature learning framework. First, we utilize the spatial second-order neighborhood in HSI to extract local information and incorporate the spectral characteristics of HSI to capture long-range information, thus constructing a reliable initial graph. Specifically, we use spatial and spectral information to construct adjacency matrices $\mathbf{A}^{\text{spa}}$ and $\mathbf{A}^{\text{spe}}$, respectively, and obtain two contrastive views (HSI spatial view and HSI spectral view) to improve the connection relationship in the graph. Second, we designed an adaptive augmentation technique that selectively removes unimportant edges and node features while highlighting important edges and node features using node centrality.

To sum up, the main contributions of this work are as follows:

1) We propose a new GCN network named UMGC for HSI classification. This network innovatively uses an unsupervised graph CL model to extract features for HSI classification.
2) We design a new multiview adjacency matrix constructor (MAMC) module to perfect the initial graph, which can obtain long-range information while focusing on the local information of nodes.
3) We design an ADAM to minimize the loss of important information caused by random augmentation and improve network performance.

The remainder of this article is organized as follows. Section II reviews the related work about GCN and CL. Section III describes the proposed UMGC method in detail. Section IV systematically elaborates on the experimental results obtained on three HSI classification benchmark datasets. Finally, Section V provides the conclusion.

II. Related Work

A. Graph Convolutional Network

GNNs follow a message-passing scheme where the node representation is learned by aggregating information from neighbors on the observed topology (that is, the initial graph structure) [40]. GCN is a typical spectral domain graph convolution that converts data from the spatial domain to the spectral domain according to graph theory and convolution theorem [47]. The convolution theorem is defined as follows:

$$f^{*}g = F^{-1}\left[F(f)\cdot F(g)\right] \tag{1}$$

where f is the input signal, g is the convolution kernel, and F^{-1} denotes the inverse Fourier transform. In the graph, the Laplace matrix in graph theory can be combined to realize the Fourier transform of the graph. The graph convolution can

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be converted as follows by using the eigenvector $\mathbf{U}$ of the Laplacian matrix as the basis function

$$x_G^* g = U\left(U^{\mathrm{T}} x \odot U^{\mathrm{T}} g\right) \tag{2}$$

where x is the input signal, $*_G$ is the graph convolution operator, and $\odot$ is a harmand product. To further simplify the expression, let $g_\theta = diag(\mathbf{U}^{\mathrm{T}} g)$ and replace the harmand product with matrix multiplication to get the convolution model as follows:

$$x_G^* g_\theta = U g_\theta U^{\mathrm{T}} x. \tag{3}$$

GCN is a special case of the above formula, which takes the first-order Chebyshev polynomials. This allows each convolution kernel to have only one learnable parameter, greatly reducing the number of parameters [34]. The final formula can be expressed as follows:

$$x_G^* g_\theta = \theta\left(\hat{\mathbf{D}}^{-1/2}\hat{\mathbf{A}}\hat{\mathbf{D}}^{-1/2}\right)x \tag{4}$$

where $\mathbf{A}$ represents the adjacency matrix of the graph, which indicates whether a connection relationship exists between each pair of nodes. $\mathbf{D}_i = \sum_j \mathbf{A}_{ij}$, where $\mathbf{D}$ stands for degree matrix, which counts the number of connections each node has with other nodes. $\hat{A} = \mathbf{A} + \mathbf{I}$ and $\hat{\mathbf{D}} = \mathbf{D} + \mathbf{I}$, where $\mathbf{I}$ represents the identity matrix. θ denotes a learnable parameter.

B. Contrastive Learning

CL has been widely used in image recognition, natural language processing, and other fields [25], [37]. The goal of CL is to learn discriminative representations by comparing positive and negative samples to capture the basic structure and relationships between different samples. Numerous algorithms have emerged in the HSI classification field. Hou et al. [37] used CL for HSI classification, which reflects the information mining ability of discriminative learning. Gao et al. [38] proposed an unsupervised meta-learning method based on multiview constraints, which combines different bands of the same sample to obtain multiview features along the spectral dimension. They differ in detail, but all contain the data augmentation [49], [50], encoder, projection head, and contrastive loss function.

CL has also shown its competitiveness and is becoming increasingly popular, which has been widely used in graph representation learning [40], [41]. Graph CL follows the mutual information maximization principle, which pulls the representation of samples with shared semantic information closer while pushing the representation of unrelated samples further away [51]. Zhu et al. [45] proposed an adaptive enhanced graph contrastive representation learning method to identify the underlying features of data at the topological and semantic levels. Yu et al. [52] first combined CL with GCN for HSI classification, utilizing a semisupervised contrastive loss function to maximize the consistency between different views of the same node or nodes belonging to the same land cover category. However, this approach is a semisupervised classification algorithm that relies, to a certain extent, on labeled samples. Therefore, we design an unsupervised CL framework to explore the interrelationships of HSI spatial–spectral features from the perspective of multiviews.

III. Proposed Method

A. Overall Framework

The architecture of the UMGC method is shown in Fig. 1. Specifically, this method mainly consists of unsupervised FE and classification. $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$ signifies the HSI, where H, W, and B are denoted as the height, width, number of bands, respectively. First, we perform PCA to reduce the dimensionality of the original HSI to minimize redundancy. The data after dimensionality reduction are represented as $\mathbf{X} \in \mathbb{R}^{H\times W\times N}$, where $N(N < B)$ represents the number of bands after PCA. Superpixels $S \in \{1, 2, \ldots, Z\}$ are then acquired through SLIC, where Z denotes the number of superpixels. The connection relationship between superpixels S can be represented by adjacency matrix $\mathbf{A}$. Thereafter, we utilize the MAMC module to construct adjacency matrices $\mathbf{A}^{\mathrm{spa}} \in \mathbb{R}^{Z\times Z}$ and $\mathbf{A}^{\mathrm{spe}} \in \mathbb{R}^{Z\times Z}$ from the spatial and spectral information views of HSI, respectively, obtaining two graphs $G^{\mathrm{spa}} = (V, \mathbf{E}^{\mathrm{spa}})$ and $G^{\mathrm{spe}} = (V, \mathbf{E}^{\mathrm{spe}})$ in the process. $\mathbf{A}$, G, V, and $\mathbf{E}$ denote the adjacency matrix, graph, nodes, and edges, respectively. $\mathbf{A}$ is an important symbol indicating whether the edges $\mathbf{E}$ in the graph G are connected. Afterward, we combine the two views with the CL framework, which can obtain topology and node features enhanced by the ADAM module to get graphs $G_{aug}^{\mathrm{spa}} = (V_{aug}, \mathbf{E}_{aug}^{\mathrm{spa}})$ and $G_{aug}^{\mathrm{spe}} = (V_{aug}, \mathbf{E}_{aug}^{\mathrm{spe}})$. UMGC extracts features by a double-layer GCN

$$\mathbf{H}^{(l+1)} = f\left(\mathbf{H}^{(l)}, \mathbf{A}\right) = \sigma\left(\hat{\mathbf{D}}^{-1/2}\mathbf{A}\hat{D}^{-1/2}\mathbf{H}^{(l)}\mathbf{W}^{(l)}\right) \tag{5}$$

where $\mathbf{H}^{(l+1)}$ represents the output embeddings of the lth layer GCN, and $\mathbf{W}^{(l)}$ represents the trainable weight matrix of the lth layer. The subsequent projection head adopts the structure of SimCLR [53]. We align the generated features, $\mathbf{Z}^{\mathrm{spa}}$ and $\mathbf{Z}^{\mathrm{spe}}$, with a contrastive loss to maximize the consistency between the two view nodes. In our framework, a symmetric normalized temperature-scaled cross-entropy loss (NT-Xent) [53] is applied

$$\zeta\left(\mathbf{z}_{\mathrm{spa},i}, \mathbf{z}_{\mathrm{spe},i}\right) = -\log \frac{e^{sim\left(\mathbf{z}_{\mathrm{spa},i}, \mathbf{z}_{\mathrm{spe},i}\right)/\tau}}{\sum_{k=1}^{n} e^{sim\left(\mathbf{z}_{\mathrm{spa},i}, \mathbf{z}_{\mathrm{spe},k}\right)/\tau}} \tag{6}$$

$$L = \frac{1}{2n}\sum_{i=1}^{n}\left[\zeta\left(\mathbf{z}_{\mathrm{spa},i}, \mathbf{z}_{\mathrm{spe},i}\right) + \zeta\left(\mathbf{z}_{\mathrm{spe},i}, \mathbf{z}_{\mathrm{spa},i}\right)\right] \tag{7}$$

where $sim(\cdot, \cdot)$ is the cosine similarity function, and τ is the temperature parameter.

In the classification stage, we adopt two classification methods. The first approach draws inspiration from node classification in graph papers [40], [41], [44] and related HSI papers [9], [37], [54]. We utilize the same backbone to fine-tune the data over 30 epochs and classify it with the softmax function. This process is referred to as UMGC_FN. The second method involves direct classification using SVM. This process is denoted as UMGC_SVM. MAMC and ADAM are described in detail below. Nomenclature summarizes some of the key symbols used in this article.

B. MAMC Module

It takes a huge amount of computing memory to convert HSI directly into a graph structure in pixels, which is difficult to

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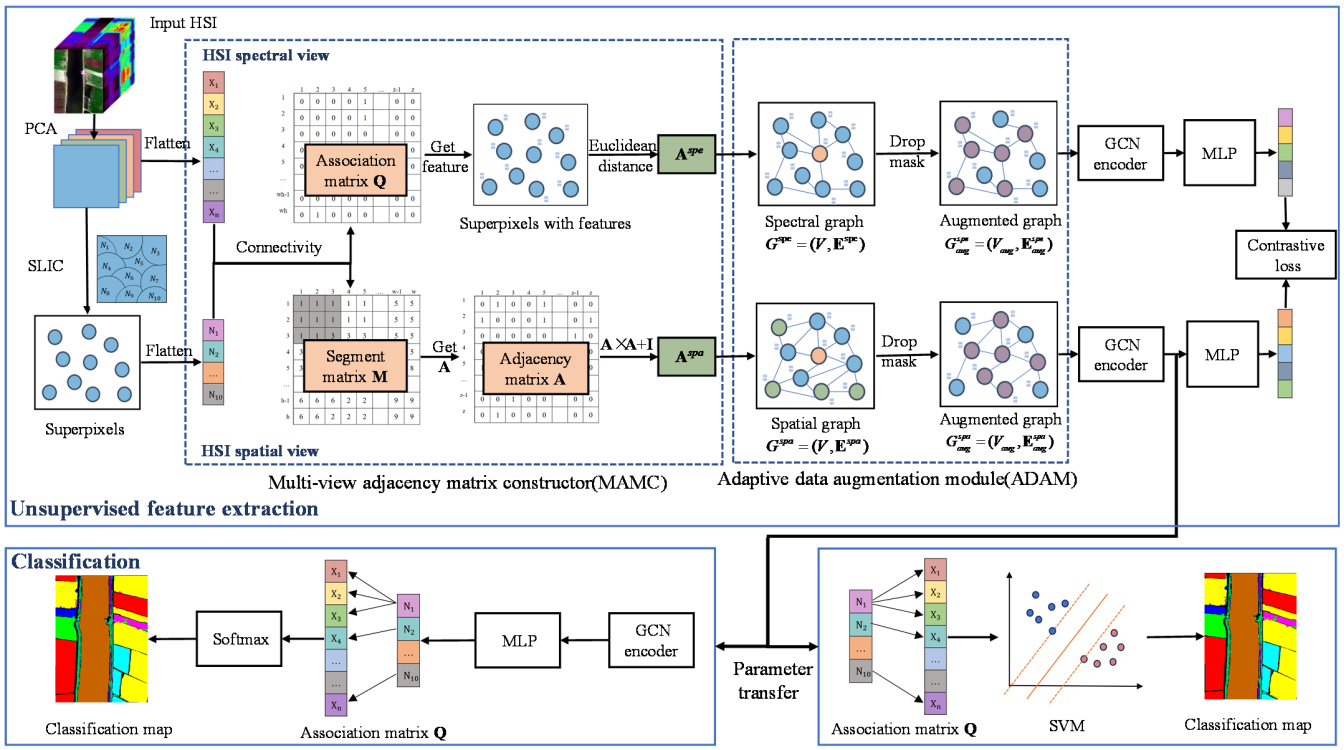


Fig. 1. Framework of the UMGC model. The UMGC framework consists of two modules: MAMC and ADAM. The nodes of the spatial second-order neighbors are shown in green. The nodes with high node centrality are shown in purple. The ordinary nodes are shown in blue. The central node is shown in orange.

perform. At present, most algorithms use the SLIC algorithm to cluster multiple pixels with similar spatial and spectral information into superpixels. SLIC is an adaptation of the k-means algorithm for superpixel segmentation, which greatly improves computing efficiency and saves computing time [31], [45]. It creates superpixels from the initial grid by iteratively growing local clusters. According to [34], each superpixel corresponds to a node

$$\mathbf{A} = \begin{cases} 1, & \text{if } S_i \text{ and } S_j \text{ are adjacent} \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

where adjacency matrix $\mathbf{A}$ is used to determine whether an edge exists between superpixels. If superpixels S_i and S_j are adjacent, then $\mathbf{A} = 1$. Otherwise, $\mathbf{A} = 0$.

However, the determination of whether two superpixels are adjacent presents an issue. The existing algorithms [31], [33], [34], [36] only consider the first-order spatial neighbors of superpixels, limiting the extraction of local information. Meanwhile, these algorithms also ignore the connection relationship of superpixels with similar spectra, which may lead to the loss of HSI long-range information. To solve this problem, we innovatively propose the MAMC module, which can find the internal correlation of spectral–spatial information from HSI to build reasonable adjacency matrixes. Subsequently, we construct multiview adjacency matrixes from the HSI spatial and spectral views.

1) *HSI Spatial View*: After the data pass through the SLIC, segment matrix $\mathbf{M} \in \mathbb{R}^{H \times W}$ is generated as follows:

$$\mathbf{M}_{ij} = S_t, \ t \in Z \tag{9}$$

where $\mathbf{M}_{ij}$ represents the pixel at the (i, j) position in $\mathbf{M}$, and S_t represents the superpixel at position $\mathbf{M}_{ij}$

$$a = \max(\gamma), \ b = \min(\gamma), \ \text{if } a \neq b, \ \mathbf{A}_{ab} = 1 \tag{10}$$

afterward we use a 3×3 window γ to traverse segment matrix $\mathbf{M}$, taking the maximum value a and the minimum value b in each γ. If $a \neq b$, then their corresponding superpixels, namely, S_a and S_b, are spatially adjacent (i.e., $\mathbf{A}_{ab} = 1$).

In this way, the adjacency matrix is established for the spatial first-order neighbors of superpixels. However, the current adjacency matrix cannot convey higher-order information. To enlarge the receptive field of the superpixel, we select the spatial second-order neighbors [55] of the superpixel. Meanwhile, we focus on the internal information of superpixel to strengthen matrix self-connection. Adjacency matrix $\mathbf{A}^{\text{spa}}$ is optimized as follows:

$$\mathbf{A}^{\text{spa}} = \mathbf{A} \times \mathbf{A} + \mathbf{I} \tag{11}$$

where $\mathbf{A}$ represents the spatial first-order adjacency matrix, and $\mathbf{I}$ represents the identity matrix.

2) *HSI Spectral View*: The SLIC will obtain an association matrix $\mathbf{Q} \in \mathbb{R}^{(H \times W) \times Z}$, which can be defined as follows:

$$\mathbf{Q}_{i,j} = \begin{cases} 1, & \text{if } ith \text{ pixel belongs to } S_j \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

where $\mathbf{Q}_{i,j}$ determines whether the ith pixel belongs to S_j. If $\mathbf{Q}_{i,j} = 1$, then the ith pixel belongs to the jth superpixel S_j

$$\mathbf{F}_S = \mathbf{Q}^{\text{T}} \mathbf{F}_P \tag{13}$$

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where $\mathbf{F}_s \in \mathbb{R}^{Z\times N}$ denotes the superpixel feature, $\mathbf{Q}^{\mathrm{T}} \in \mathbb{R}^{Z\times(H\times W)}$ denotes the transpose of the association matrix, and $\mathbf{F}_P$ denotes the pixel feature. The generation of the spectral adjacency matrix is mainly based on the Euclidean distance of the spectrum. For each node $\mathbf{v}$ of $\mathbf{F}_s$, we calculate its Euclidean distance from all other nodes. Then, we select the K-nearest neighbors to generate the spectral adjacency matrix, which helps in connecting the HSI long-range information

$$N(\mathbf{v}_i) = \arg sort\left(||\mathbf{v}_i - \mathbf{v}_j||^2\right) \tag{14}$$

$$\mathbf{A}^{\mathrm{spe}} = \begin{cases} 1, & \text{if } \mathbf{v}_j \in N_K(\mathbf{v}_i) \\ 0, & \text{otherwise} \end{cases} \tag{15}$$

where $\arg sort()$ is a sorting function that arranges the values from smallest to largest and outputs the corresponding array subscript. $N(\mathbf{v}_i)$ means to calculate the Euclidean distance from the ith node $\mathbf{v}_i$ to all other nodes $\mathbf{v}_j$. $N_K(\mathbf{v}_i)$ means the K-nearest spectral neighbors of the current node $\mathbf{v}_i$. Finally, the long-range information of the graph structure is connected by ordering the Euclidean distance of each node. The default value $K = 16$ is taken after we have conducted relevant experiments in combination with [29].

In summary, we construct a multiview adjacency matrix to fully use HSI from the spatial and spectral perspectives. The specific process is shown in Fig. 1.

C. Adaptive Data Augmentation Module

Data augmentation schemes have been proven to be a key component of visual representation learning [41]. The majority of the current HSI CL methods randomly augment the input images and maximize the consistency of the two branches. Most of these methods are specifically designed for pixel-level classification tasks [37], [38], [56]. Accordingly, we aim to design an adaptive graph enhancement module that retains important features and edges and discards unimportant and insensitive representations according to the importance principle of nodes compared with [45]. Wu et al. [57] indicated that the combination of multiple data augmentation operations is critical in defining a contrastive prediction task. Thus, for the unsupervised node classification task, we design an adaptive augmentation method from the edge and node feature levels.

For the edge level, we use node centrality as an evaluation metric for node importance. Node centrality is a widely used metric for quantifying the influence of nodes in a graph [45]. If the two nodes adjacent to an edge are important enough, then the edge is also important. This mechanism will selectively remove unimportant edges and keep critical ones

$$\mathbf{W}^e_{ij} = \left(\vartheta_i(\mathbf{v}) + \vartheta_j(\mathbf{v})\right)/2. \tag{16}$$

We define the centrality of an edge $\mathbf{W}^e_{ij}$ as the average of the centrality scores of two adjacent nodes. $\vartheta_i(\mathbf{v})$ and $\vartheta_j(\mathbf{v})$ represent the node centrality for node i and j. e is the short for edge

$$idx^e = \arg sort\left(\frac{\mathbf{w}^e_{\max} - \mathbf{w}^e}{\mathbf{w}^e_{\max} - \mathbf{w}^e_{\mathrm{mean}}}\right). \tag{17}$$

Given that edge centrality values may vary across orders of magnitude, we use a normalization step to simplify the values of edge centrality. idx^e represents the array subscript of all edge weights $\mathbf{W}^e$ output by $\arg sort()$. $\mathbf{W}^e_{\max}$ and $\mathbf{W}^e_{\mathrm{mean}}$ are the maximum and average of $\mathbf{W}^e$.

Depending on the size of the center value, we can remove the edge with the lower score. To preserve the important edges, the formula can be written as

$$\mathbf{E}_i = \begin{cases} 0, & \text{if } i \in idx^e[0, \mathrm{T}_1] \\ 1, & \text{if } i \in idx^e[\mathrm{T}_1, J]. \end{cases} \tag{18}$$

Equation (18) is used to determine whether the ith edge $\mathbf{E}_i$ should be removed. If edge $\mathbf{E}_i$ is zero, then it should be removed. If $\mathbf{E}_i$ is one, then we keep it. In this case, we will count the number of removed edges T_1. $\mathrm{T}_1 = J \times p_1$, where $p_1 \in [0, 1]$ is a hyperparameter that controls the discard probability of the edge. J represents the edge number. Accordingly, we can get an adaptive augmentation edge vector.

In feature augmentation, we augment node attributes by masking some dimensions of nodes [45]. We think node features are closely related to the central node and assume that feature dimensions that frequently occur on influential nodes should be important. Similar to the edge augmentation method, the weight of node features is obtained by calculating as follows:

$$\mathbf{W}^f = \mathbf{X}^{\mathrm{T}}\vartheta(\mathbf{v}) \tag{19}$$

where $\vartheta(\mathbf{v})$ is a node centrality measure that is used to quantify node importance, $\mathbf{X}^{\mathrm{T}}$ is the transpose of the input feature, $\mathbf{W}^f$ is the feature weight of the node, and f is short for feature. Similarly, we use normalization to simplify the values of feature centrality

$$idx^f = \arg sort\left(\frac{\mathbf{w}^f_{\max} - \mathbf{w}^f}{\mathbf{w}^f_{\max} - \mathbf{w}^f_{\mathrm{mean}}}\right) \tag{20}$$

where $\mathbf{W}^f_{\max}$ and $\mathbf{W}^f_{\mathrm{mean}}$ are the maximum and average of $\mathbf{W}^f$, respectively. Depending on the size of the center value, we can remove the feature with the lower score. The formula is as follows:

$$\mathbf{F}_i = \begin{cases} 0, & \text{if } i \in idx^f[0, \mathrm{T}_2] \\ \mathbf{F}_i, & \text{if } i \in idx^f[\mathrm{T}_2, N]. \end{cases} \tag{21}$$

Equation (21) is used to determine whether the ith feature $\mathbf{F}_i$ should be removed. The T_2 value is calculated in the same way as edge augmentation. $\mathrm{T}_2 = N \times p_2$, where $p_2 \in [0, 1]$ is a hyperparameter that controls the discard probability of the feature, and N represents the feature dimension after PCA.

Finally, we perform data augmentation at the edge and feature level by combining the HSI spatial–spectral views. The adaptive augmentation process for the multiview is shown in Fig. 2.

IV. Experimental Results

A. Data Description

In order to comprehensively evaluate the classification performance of UMGC, we adopted three benchmark HSI datasets. The three datasets are described in detail below:

1) *Salinas*: The Salinas dataset was taken with the airborne visible-infrared imaging spectrometer (AVIRIS) sensor

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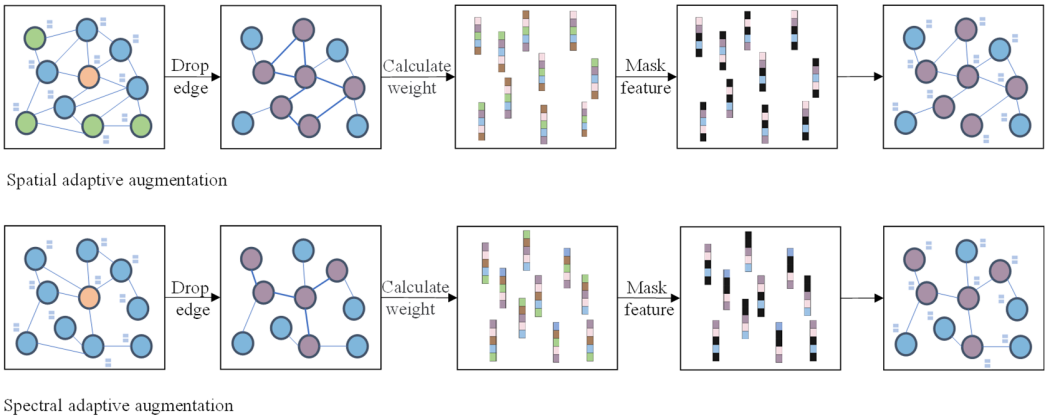


Fig. 2. Flowchart of the ADAM. The nodes of the spatial second-order neighbors are shown in green. The nodes with high node centrality are shown in purple. The ordinary nodes are shown in blue. The central node is shown in orange. The masked features are shown in black.

TABLE I

NUMBER OF TRAINING AND TEST SAMPLES FOR EACH CLASS OF THE SALINAS DATASET

No.	Color	Class	Train	Test
1		Broccoli-green-weeds-1	21	1988
2		Broccoli-green-weeds-2	38	3688
3		Follow	20	1956
4		Fallow-rough-plow	14	1380
5		Fallow-smooth	27	2651
6		Stubble	40	3919
7		Celery	36	3543
8		Grapes-untrained	113	11158
9		Soil-vineyard-develop	63	6140
10		Corn-weeds	33	3245
11		Lettuce-romaine-4wk	11	1057
12		Lettuce-romaine-5wk	20	1907
13		Lettuce-romaine-6wk	10	906
14		Lettuce-romaine-7wk	11	1059
15		Vinyard-untrained	73	7195
16		Vinyard-vertical-trellis	19	1788
		Total	549	53580

TABLE II

NUMBER OF TRAINING AND TEST SAMPLES FOR EACH CLASS OF THE WHU-HI-LONGKOU DATASET

No.	Color	Class	Train	Test
1		Corn	173	34338
2		Cotton	42	8332
3		Sesame	16	3015
4		Broad-leaf soybean	317	62895
5		Narrow-leaf soybean	21	4130
6		Rice	60	11794
7		Water	336	66720
8		Roads and houses	36	7088
9		Mixed weed	27	5202
		Total	1028	203514

in Salinas Valley, California [58]. The dataset size is 512 × 217 and contains a total of 111 104 pixels. It contains 224 continuous bands within 0.36–2.5 μm, and the spatial resolution is 3.7 m. After removing 20 absorbent bands (108–112, 154–167, 224), 204 bands are used for training. There are 16 types of surface objects in the study area, including vegetables, bare soils, vineyard fields, corn weeds, etc. Table I lists the numbers of the training and test samples for classification.

2) *WHU-Hi-LongKou*: The Wuhan University hyperspectral image (WHU-Hi) dataset was acquired in WHU-Hi-LongKou town, China in 2018, with an unmanned aerial vehicle (UAV) platform [59]. It can be used as the benchmark dataset for HSI crop classification. The dataset size is 550 × 400. The study area is a simple agricultural scenario containing nine classes: corn, cotton, sesame, broad-leaf soybean, narrow-leaf soybean, rice, roads, houses, water, and mixed weeds. The flight height of the UAV is 500 m, and the spatial resolution is about 0.463 m. There are 270 bands within 0.4–1 μm. Table II lists the numbers of the training and test samples for classification.

3) *Trento*: The Trento dataset was taken in a rural area south of the city of Trento, Italy [60]. The HSI data were collected with the AISA Eagle sensor, which consisted of 600 × 166 pixels with 63 spectral channels and a range of 0.40–0.98 μm. The spatial resolution is 1 m. In this dataset, six classes are included: apple trees, buildings, ground, woods, vineyards, and roads. Table III lists the numbers of the training and test samples for classification.

TABLE III

NUMBER OF TRAINING AND TEST SAMPLES FOR EACH CLASS OF THE TRENTO DATASET

No.	Color	Class	Train	Test
1		Apple trees	41	3993
2		Buildings	30	2873
3		Ground	5	474
4		Woods	92	9031
5		Vineyard	106	10395
6		Roads	32	3142
		Total	306	29908

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TABLE IV
COMPARISON RESULTS OBTAINED ON THE SALINAS DATASET

Class	EMP-SVM	AAE	VAE	DMVL	SMF-UL	UMGC_SVM	UMGC_FN
Broccoli-green-weeds-1	99.20	99.54	99.03	99.15	96.33	98.87	**100.00**
Broccoli-green-weeds-2	99.32	99.86	99.92	95.88	97.29	**100.00**	**100.00**
Follow	46.11	97.99	**100.00**	**100.00**	94.27	**100.00**	**100.00**
Fallow-rough-plow	97.54	97.14	**99.63**	95.10	98.84	94.57	90.43
Fallow-smooth	97.96	**100.00**	99.28	93.33	92.38	95.70	97.36
Stubble	98.88	99.28	99.54	98.15	94.21	98.14	**99.80**
Celery	99.49	**99.89**	98.75	97.33	89.90	99.10	99.69
Grapes-untrained	98.40	97.51	97.11	95.05	97.43	**99.68**	95.64
Soil-vineyard-develop	99.46	**100.00**	**100.00**	99.84	95.93	**100.00**	99.92
Corn-weeds	87.40	96.98	94.02	**100.00**	87.80	91.49	96.02
Lettuce-romaine-4wk	0.00	99.90	**100.00**	98.97	94.70	93.38	93.85
Lettuce-romaine-5wk	**100.00**	99.74	75.69	97.66	89.09	89.15	91.98
Lettuce-romaine-6wk	0.55	**99.55**	98.22	94.41	88.52	95.25	95.47
Lettuce-romaine-7wk	90.37	96.37	80.82	86.95	84.89	**97.73**	84.04
Vinyard-untrained	0.00	14.32	0.00	80.96	92.72	95.96	**99.25**
Vinyard-vertical-trellis	78.69	98.70	99.04	**100.00**	87.57	94.57	99.89
OA(%)	78.53	87.43	84.10	94.86	90.63	95.81	**97.53**
AA(%)	74.59	93.54	90.06	95.80	92.61	95.85	**96.46**
kappa×100	75.74	85.87	82.14	94.27	86.95	95.33	**97.25**

TABLE V
COMPARISON RESULTS OBTAINED ON THE WHU-HI-LONGKOU DATASET

Class	EMP-SVM	AAE	VAE	DMVL	SMF-UL	UMGC_SVM	UMGC_FN
Corn	98.37	99.01	95.37	98.28	97.43	**99.69**	99.48
Cotton	71.44	89.33	95.79	99.53	88.31	93.73	**99.64**
Sesame	0.00	98.21	**98.61**	43.71	79.14	90.38	97.74
Broad-leaf soybean	98.12	97.00	**98.69**	94.33	94.17	98.08	97.55
Narrow-leaf soybean	0.00	77.27	63.56	32.42	66.66	84.00	86.39
Rice	94.37	96.62	89.06	90.93	97.75	**99.67**	99.64
Water	**99.97**	89.13	98.59	99.19	99.13	99.89	99.50
Roads and houses	90.04	81.75	66.03	81.55	81.96	**91.13**	89.70
Mixed weed	5.40	**92.28**	77.70	84.28	61.45	72.57	47.15
OA(%)	91.36	93.39	95.03	93.90	91.96	**97.56**	96.94
AA(%)	61.96	91.17	87.04	80.46	85.11	**92.13**	90.76
kappa×100	88.46	91.32	93.44	91.95	89.75	**96.79**	95.98

TABLE VI
COMPARISON RESULTS OBTAINED ON THE TRENTO DATASET

Class	EMP-SVM	AAE	VAE	DMVL	SMF-UL	UMGC_SVM	UMGC_FN
Apple trees	80.34	80.83	**99.30**	84.14	86.54	93.49	97.40
Buildings	80.61	42.69	86.56	**87.80**	76.06	73.16	85.42
Ground	**95.15**	77.54	16.03	45.84	53.86	79.54	92.83
Woods	98.79	99.36	99.64	99.71	91.29	99.84	**99.87**
Vineyard	98.76	99.61	99.72	96.35	94.95	**99.94**	98.88
Roads	72.76	**94.70**	48.31	81.41	67.86	80.74	74.76
OA(%)	91.78	91.02	91.84	92.54	92.66	94.14	**95.06**
AA(%)	87.73	82.63	75.03	82.54	78.43	87.79	**91.53**
kappa×100	89.96	87.91	89.08	90.00	79.21	92.11	**93.42**

B. Experimental Setup

In this section, we first introduce the detailed structure of our proposed UMGC and the parameter settings of the experiments. Then, we show all the baseline models used for comparison and the settings of their parameters.

1) *Implementation Details*: During the unsupervised FE phase, we carefully tuned multiple hyperparameters to ensure that the model could learn features efficiently and accurately from the data. Specifically, we set the number of training epochs to 200, the hidden dimensions to 128, and the learning rate to 0.005 and selected Adam as the optimizer. In the contrastive loss function, the tau value is set to 0.2. Given that we used the entire image for FE, the batch size is set to one. Through hyperparameter experiments, the PCA dimensionality reduction band numbers for the Salinas, LongKou, and Trento datasets are set to 35, 10, and 15, respectively. The superpixel scales are set to 40, 35, and 30.

Furthermore, we use double-layer GCN as the encoder and Linear-BN-ReLU-Linear as MLP. After the unsupervised FE, we use the extracted features for supervised classification. The 0.1%, 0.05%, and 0.1% samples are randomly selected for training the Salinas, LongKou, and Trento datasets. We select

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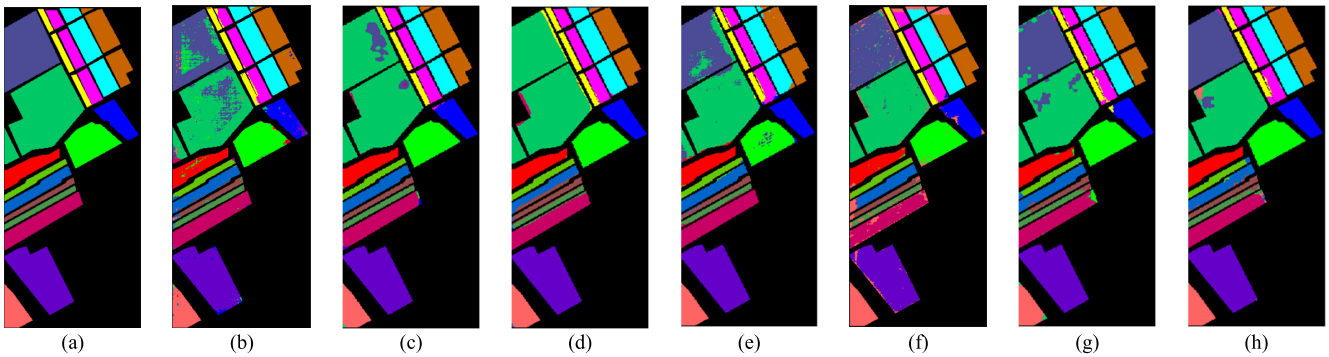


Fig. 3. Classification maps of different methods on the Salinas dataset. (a) Ground truth map. (b) EMP-SVM. (c) AAE. (d) VAE. (e) DMVL. (f) SMF-UL. (g) UMGC_SVM. (h) UMGC_FN.

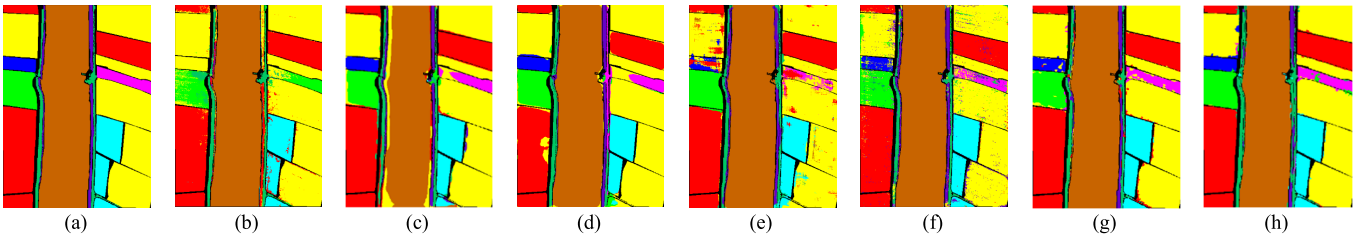


Fig. 4. Classification maps of different methods on the WHU-Hi-LongKou dataset. (a) Ground truth map. (b) EMP-SVM. (c) AAE. (d) VAE. (e) DMVL. (f) SMF-UL. (g) UMGC_SVM. (h) UMGC_FN.

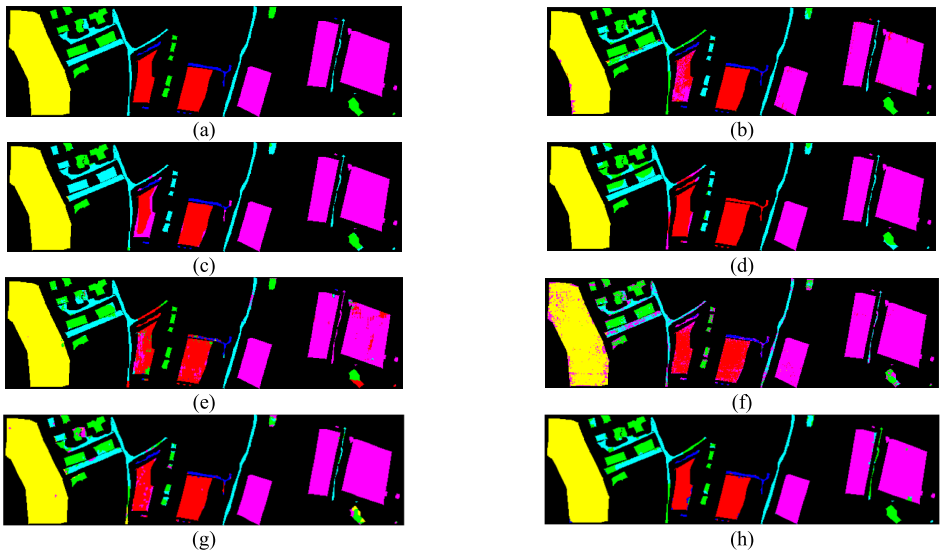


Fig. 5. Classification maps of different methods on the Trento dataset. (a) Ground truth map. (b) EMP-SVM. (c) AAE. (d) VAE. (e) DMVL. (f) SMF-UL. (g) UMGC_SVM. (h) UMGC_FN.

samples by rounding them up to prevent the training set from being empty.

Six quantitative evaluation indicators are used in our experiments, including per-class accuracy, overall accuracy (OA), average accuracy (AA), kappa coefficient (kappa), unsupervised FE training time (FE training), and classification training time (classification). We also provide a visualization of the experimental results.

All experimental results are the median of five independent runs. The proposed algorithm is implemented via Python 3.10.11 and PyTorch 1.11.0. The hardware used for training is an i7-10700K CPU and an NVIDIA GeForce RTX 3090 GPU.

2) *Comparison Methods*: To evaluate the classification performance of UMGC, we select five unsupervised methods for comparison, including a traditional EMP-SVM [18] method and four CNN-based unsupervised learning methods, i.e., adversarial autoencoder (AAE) [61], which uses the idea of adversarial to limit the distribution of potential code by training generative adversarial network (GAN) [62]. Variational autoencoder VAE [61] used KL-divergence and the reparameterization trick to restrict latent code distribution.

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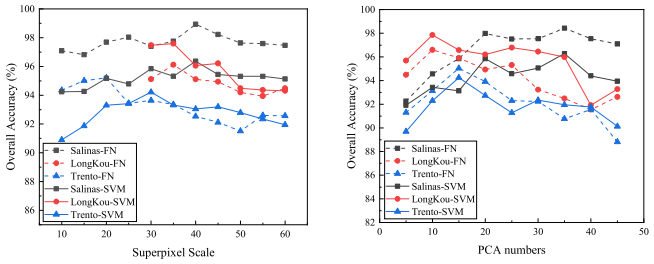


Fig. 6. Classification performance of different parameters on three datasets on UMGC_SVM and UMGC_FN. (a) Segmentation scales. (b) PCA numbers.

Deep multiview learning (DMVL) [63] trains the network to learn view-invariant features. Unsupervised spectral motion feature learning (SMF-UL) [64] reinterprets the 3-D HSI data from the perspective of spectral sequence and attempts to extract the spectral variation information as the spectrum motion feature. All comparison models are performed in the same environment using the original paper's hyperparameters setting. Given that the classifiers of these methods are all SVM, the results of the comparison algorithm can have a more intuitive comparison with the UMGC_SVM. Meanwhile, we choose the radial basis function (RBF) kernel by default for SVM.

C. Classification Results

Tables IV–VI present detailed classification accuracies of the different methods on three datasets. The results demonstrate that our UMGC method, leveraging the initial graph structure constructed by the MAMC module and CL framework, exhibits superior deep spectral–spatial FE capabilities. Compared to other comparison algorithms, UMGC_FN and UMGC_SVM achieved the optimal and second-best OA on all three datasets. Specifically, the traditional EMP-SVM method exhibits limitations in FE, making it difficult to distinguish ground object categories with similar spectral characteristics accurately. For instance, in the Salinas dataset, EMP-SVM completely misclassified the lettuce-romaine-5wk and vinyard-untrained categories, assigning them an accuracy of 0. While CL methods based on DL have improved OA, they still show deficiencies in identifying certain specific categories. For example, in the Salinas dataset, the recognition rate for the vinyard-untrained category is relatively low. In the LongKou dataset, the narrow-leaf soybean category is not well identified. And, in the Trento dataset, the ground category also faces challenges in recognition. In contrast, our UMGC method can accurately distinguish object categories with similar spectral characteristics, demonstrating excellent performance in identifying various categories and thus achieving higher classification accuracy.

The qualitative evaluation results of the different methods are shown in Figs. 3–5. The intraclass variability and interclass similarity of the dataset make the FE challenging, resulting in significant differences in per-class accuracy and OA among the six algorithms. Our UMGC_SVM and UMGC_FN demonstrated excellent performance in distinguishing between purple and green blocks in the Salinas dataset, verifying the

TABLE VII

OA (%) INDICES OF THE COMBINATION OF DIFFERENT MODULES ON THREE DATASETS

Baseline FN	Random drop	Adaptive drop	MAMC	OA		
				Salinas	LongKou	Trento
✓				96.48	92.99	94.74
✓			✓	97.35	94.53	95.04
✓		✓		97.47	93.81	93.54
✓	✓		✓	97.95	95.59	95.26
✓		✓	✓	**98.39**	**96.96**	**95.75**

TABLE VIII

OA (%) INDICES OF THE COMBINATION OF DIFFERENT MODULES ON THREE DATASETS

Baseline SVM	Random drop	Adaptive drop	MAMC	OA		
				Salinas	LongKou	Trento
✓				95.10	96.07	92.47
✓			✓	95.18	96.26	93.84
✓		✓		95.86	97.63	93.52
✓	✓		✓	95.52	97.76	92.98
✓		✓	✓	**96.65**	**97.97**	**94.23**

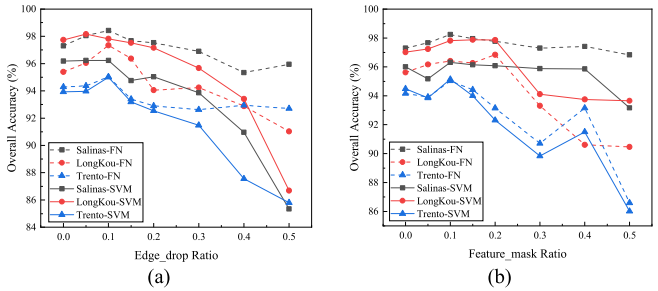


Fig. 7. Classification performance of different edge_drop ratios and feature_mask ratios on three datasets on UMGC_SVM and UMGC_FN. (a) Edge_drop ratio. (b) Feature_mask ratio.

effectiveness of our FE approach. In the LongKou dataset, the traditional EMP-SVM algorithm cannot recognize the magenta block when the labeled samples are limited. DL algorithms have many misclassified blocks and fail to achieve satisfactory classification results. In Fig. 5, the Trento dataset poses challenges for classification algorithms due to uneven sample quantities and varying category shapes. Comparison algorithms struggle to accurately classify the categories of slender distributions. However, the advantage of the UMGC method lies in its full utilization of the local and long-range information and the graph's information transfer structure. This comprehensive feature learning ability enables UMGC to better capture the potential features of the data in handling HSI classification problems, thereby improving classification accuracy.

D. Discussions

1) *Ablation Study*: To further highlight the impact of key modules in the UMGC (MAMC and ADAM), we conducted multiple ablation studies on three datasets. Tables VII and VIII show the OA accuracy of the three datasets under different module combinations, where the highest accuracy is indicated in bold. Baseline means building a graph structure using only

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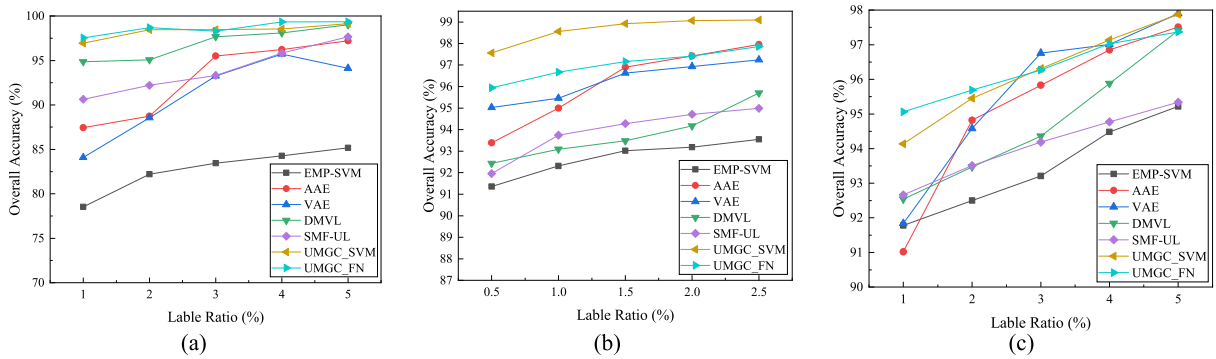


Fig. 8. Classification performance of each method with different training label ratios on three datasets. (a) Salinas. (b) WHU-Hi-LongKou. (c) Trento.

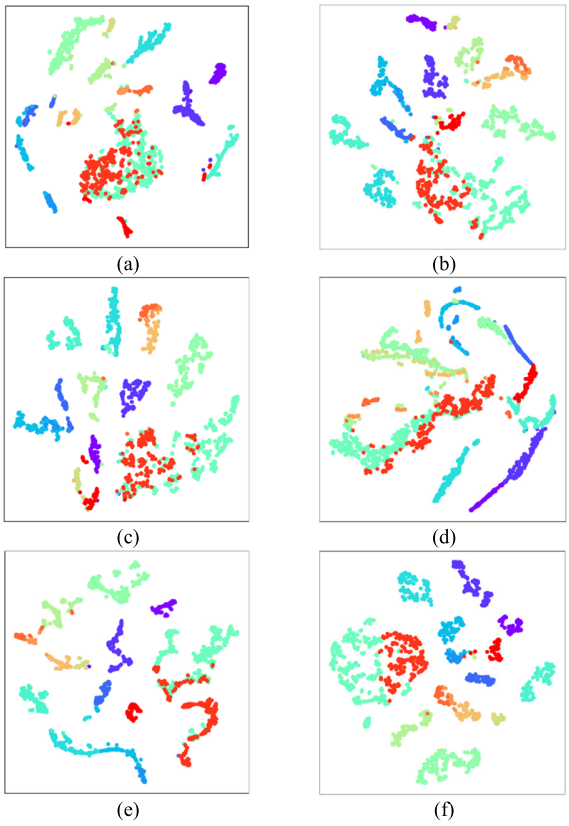


Fig. 9. T-SNE feature visualization of different comparison methods on the Salinas dataset. (a) EMP-SVM. (b) AAE. (c) VAE. (d) DMVL. (e) SMF-UL. (f) UMGC.

spatial information and feeding the graph into two identical GCN branches.

Tables VII and VIII show that UMGC demonstrates superior accuracy when only using the Baseline FN and Baseline SVM compared with the other algorithms. This feature is attributed to the framework of CL, which enables the extraction of advanced feature information of data. In the three datasets of the FN branch built with MAMC, the improvements compared with the baseline are 0.87%, 2.52%, and 0.3%, and those in the SVM are 0.08%, 0.19%, and 0.37%. This result is obtained because the use of Euclidean distance to build spectral links in the initial adjacency matrix can enhance the model's generalization ability. The experimental results show that adaptive drop can significantly increase by 1%–2% compared with random drop when all MAMC is used. This situation arises because adaptive drop can selectively discard unimportant edge and node features to avoid the loss of key information. Specifically, the combination of the two modules improved the OAs on the Salinas, LongKou, and Trento datasets by 1.91%, 3.97%, and 1.01%, respectively, compared with the baseline on the FN branch. In the SVM branch, the OAs on the Salinas, LongKou, and Trento datasets increased by 1.55%, 1.9%, and 1.75%, respectively.

2) *Parameter Sensitivity Analysis*: We utilize three datasets to quantitatively analyze four hyperparameters: the superpixel scale, the number of PCA, edge_drop ratio, and feature_mask ratio.

a) Superpixel scale: Fig. 6(a) shows an analysis of the superpixel scale. The segmentation of superpixels is crucial to the initial graph. Too large segmentation areas may contain multiple categories, resulting in confusion about superpixel information. Too small segmentation areas will increase the computing burden of the computer, making the memory insufficient. Salinas is relatively stable and reaches the optimal accuracy on both branches at an area of 40. LongKou also reached its highest accuracy at 35. The LongKou dataset is relatively large. No relevant precision is observed when the scale is less than 30 due to computational memory limitations. The most partition areas of the Trento dataset on the two branches of FN and SVM are 20 and 30. The experiments demonstrate that the diverse algorithms of the different datasets exhibit varying preferences in the selection of superpixel segmentation scales.

b) PCA numbers: Fig. 6(b) shows an analysis of the number of bands in PCA. PCA can reduce data redundancy and remove noise. However, a considerable number of compressed bands can result in information loss. Our algorithm UMGC needs certain bands to generate a spectral adjacency matrix by extracting the information. In the ADAM, a certain number of bands are also required to make partial feature masks. Accordingly, a discussion on how to find a suitable number of bands on the three datasets is relevant. The picture shows that the overall trend of the same dataset in FN and SVM is roughly the same. The classification accuracy of SVM is better than that of FN in the LongKou, which may be because the

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TABLE IX
COMPARISONS IN TERMS OF THE PRETRAINING TIME AND TRAINING TIME OF DIFFERENT DL-BASED METHODS

Dataset	Time	EMP-SVM	AAE	VAE	DMVL	SMF-UL	UMGC _SVM	UMGC _FN
Salinas	FE training(m)	**0.27**	116.33	40.65	103.34	76.45	0.58	0.58
	Classification(s)	4.93	81.03	78.38	21.56	42.41	**2.46**	6.40
WHU-Hi-LongKou	FE training(m)	**0.42**	110.13	43.27	226.71	76.45	0.48	0.48
	Classification(s)	16.91	233.68	82.64	35.73	143.19	13.52	**5.63**
Trento	FE training(m)	**0.25**	59.56	37.88	241.66	76.45	1.41	1.41
	Classification(s)	3.13	25.72	25.01	11.05	137.39	**2.72**	14.30
Trainable parameters(M)		\	0.76	0.82	11.31	3.55	**0.58**	**0.58**
FLOPs(G)		\	1.61	3.23	58.72	17.84	**0.18**	**0.18**

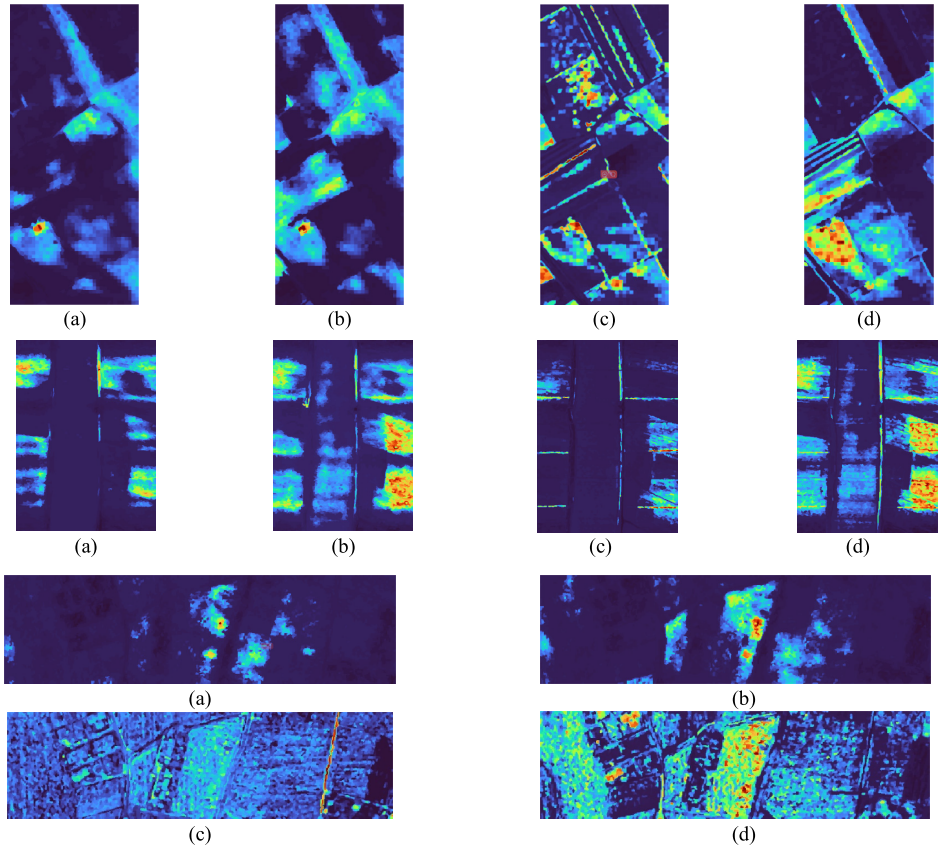


Fig. 10. Visual comparison of features under different MAMC combinations in Salinas, LongKou, and Trento datasets, respectively. (a) Spatial first-order neighbors. (b) Spatial second-order neighbors based on self-connection. (c) Spectral information. (d) Spatial second-order neighbors based on self-connected and spectral information.

LongKou dataset is relatively simple, and enough features have been extracted during pretraining. However, FN will further learn the features, leading to data overfitting and accuracy reduction. The optimal band numbers of the Salinas, LongKou, and Trento datasets are 35, 10, and 15, respectively.

c) Edge_drop ratio and feature_mask ratio: Fig. 7 shows an analysis of the ratio of adaptive augmentation. The efficiency of classification will significantly decrease when the discard ratio is too high for edge and nodes. This situation is attributed to the high drop ratio that can result in excessive damage to the graph structure, making it difficult for GNN to learn useful information from the node's neighborhood. FN and SVM show roughly the same trend in the same dataset, indicating that the loss of node features has a great influence on the graph structure. The probability of augmentation of our UMGC on these three datasets is around 0.1, indicating that the appropriate graph perturbation helps in enhancing the robustness of the model.

3) *Model Complexity and Efficiency Analysis*: We analyze the FE training time and the classification time of each algorithm under optimal classification accuracy. As shown in Table IX, the minimum time is highlighted in bold font, and

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the subminimum value is underlined. In DL, the training time of GNNs is shorter than that of CNNs, because the existence of superpixels greatly reduces the computing memory and improves the computing speed. Our pretraining time is second only to the traditional machine learning algorithm EMP-SVM and far ahead of other deep learning algorithms. Furthermore, the number of trainable parameters and the floating point of operations (FLOPs) [34] are adopted to measure the model complexity of the different DL-based methods. The results are shown in Table IX. The number of trainable parameters and FLOPs in UMGC are about 0.58 M and 0.18 G, which are less than those of the other comparison methods. This result indicates that UMGC converts numerous pixels into fewer superpixels, which requires less computation than the contrast algorithm.

4) *Influence of Label Ratio*: In most cases, labeling samples of HSI data is expensive and requires a substantial amount of manpower and resources. Therefore, we hope that after unsupervised FE, UMGC can achieve better classification accuracy with few labeled samples. Fig. 8 shows that our UMGC achieves the best accuracy on all three datasets, with OA accuracy increasing with the increase in the sample size. Furthermore, the OA gap between algorithms is evident when the sample size is low. The accuracy gap gradually decreases with the increase in the sample size. In conclusion, the UMGC algorithm proposed in this work is superior to other algorithms in the case of small samples, demonstrating the effectiveness of pretraining to extract features.

5) *Visualization Analysis*: In order to compare the FE performance of different unsupervised algorithms, we use the t-stochastic neighbor embedding (t-SNE) technique to visualize the data distribution of Salinas in the learning feature space [65]. Approximately 5% of the Salinas samples are taken for visualization for visualization. As can be seen from Fig. 9(a), compared to other methods, our UMGC exhibits better separability and fewer misclassifications in each category. Specifically, the separability of cyan and red is significantly improved by the UMGC method compared with the other algorithms, further showing that it can significantly maximize the interclass variance and minimize the intraclass variance.

To more intuitively reflect our MAMC model for local and long-range information extraction ability, we visualize the features before the projection head in the Salinas, LongKou, and Trento datasets, as shown in Fig. 10. Fig. 10(a) depicts a feature map using only spatial first-order neighbors, which has a limited FE range. Fig. 10(b) shows a second-order neighbor in space based on self-attention. This neighbor has a larger sensory area and focuses on more information than that in Fig. 10(a). Fig. 10(c) depicts a feature map using only a spectral adjacency matrix, which can extract the long-range information of the entire map and better distinguish the boundary of ground objects. However, relying solely on spectral information may not adequately capture the important details. Fig. 10(d) is a combination of Fig. 10(b) and (c) to highlight key information about the immediate neighborhood while capturing global information. This notion indicates that the spatial–spectral information of HSI must be fully considered when constructing the initial graph structure. In summary, the feature maps of the three datasets in Fig. 10 visually illustrate the advantages of the MAMC model.

E. Limitations of Method

Previous experiments have fully demonstrated the remarkable performance of our UMGC in FE. However, as with any technology, our method also has some limitations that need to be pointed out. For instance, given the challenges that GCN models may encounter when deeply stacked, we chose a two-layer GCN architecture to avoid the over-smoothing issue and ensure the model's performance. Furthermore, when constructing adjacency matrix A, we may encounter a small number of incorrect node connections in cases where the geographical features are densely and chaotically distributed in small areas. These issues limit the applicability of our method in handling such data. Accordingly, our method is suitable for datasets with relatively regular geographical feature distributions and manageable sizes. We will endeavor to improve and optimize our method through further research and technological advancements to effectively address various complex scenarios and datasets.

V. Conclusion

This study proposes an unsupervised multiview graph CL method for HSI classification. This method reconstructs the connection relationship between pixels and superpixels to optimize the initial graph structure. We also improve the robustness of the model through an ADAM. First, we introduce the concept of multiview and construct the adjacency matrix of superpixels from the HSI spatial and spectral views, which can extract the local and long-range information of HSI. Second, we adaptively remove edges and mask features according to the principle of node centrality to improve the robustness of the CL framework. Finally, we divide the obtained features into two classification branches. One maps superpixel pixels to SVM for direct classification, and the other uses GCN for fine-tuning and classification. A large number of experiments on three widely used HSI datasets demonstrate the validity and rationality of our proposed UMGC approach.

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